

ONE MODEL, MANY SKELETONS: SKELETON-AWARE IN-CONTEXT TEXT-TO-MOTION DIFFUSION ACROSS HETEROGE- NEOUS TOPOLOGIES

Anonymous authors

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ABSTRACT

Text-to-motion generation has mostly been built for one body: the joint count, joint order and pose parameterisation of a single human skeleton are baked into the architecture. We present TOPX, a text-conditioned motion diffusion transformer that serves hundreds of heterogeneous animal skeletons with one set of weights. Three ingredients make this work. KTJD-17 is a per-joint representation that any rigged animation can be converted to from its kinematic tree, rest pose and motion alone, with every (rig, joint, channel) cell normalised by its own statistics, so that rigs of 34 to 102 joints share one network input without a template, retargeting or inverse kinematics. The skeleton enters as a prompt: its rest pose is prepended to the sequence as a clean frame, and natural-language descriptions of its joints together with a graph-distance attention bias tell the network what each joint is; nothing in the model is specific to a skeleton, so a rig it has never seen is served by the same forward pass once its statistics are supplied. And a grouped flow-matching objective whose channel weights are calibrated from measured group energies rather than hand-tuned keeps root, body and contact channels balanced across rigs of very different size. We build a captioned corpus of 311 production animal rigs (77,894 clips), train a 303M-parameter model on it, and score it with a skeleton-aware text–motion evaluator on 3,899 held-out clips (R@1 0.922 against 0.964 for the real clips, FID 0.005). On the Truebones zoo, rigs outside the library, a per-rig low-rank adapter trained on about ten clips brings the model’s rotation and position channels back into agreement on bodies far outside the training distribution, and we report descriptive kinematic statistics of the outputs next to the released AnyTop model on the same rigs.

1 INTRODUCTION

Text-to-motion generation has made striking progress, but almost all of it on one body. The standard pipeline—a diffusion model, a motion tokenizer or a latent generator trained on HumanML3D (Guo et al., 2022)—assumes a fixed kinematic tree, typically the 22-joint SMPL skeleton, and bakes it into every layer: the input projection is sized for that joint count and any codebook is learned for one morphology. The result is a family of models that animate a human well and cannot animate anything else.

The applications that most need generative motion—games, film previsualisation, virtual production—are populated by characters that are not human, and a studio does not have one skeleton but hundreds, each authored by a rigger with its own joint count, naming scheme and helper bones. Our corpus is a faithful sample of that reality: 311 production animal rigs with 34 to 102 deformation joints (median 64), 108 distinct kinematic trees and 77,894 captioned clips (6.2M frames).¹ Generating on a canonical skeleton and *retargeting* (Villegas et al., 2018; Aberman et al., 2020; Lee

¹The clips and their captions follow the AniMo4D annotation set of Wang et al. (2025b); the rigs are re-exported with deformation bones only, the joint descriptions are ours, and the conversion to KTJD-17 is new (App. A). The Truebones zoo, a publicly distributed set of 66 animal rigs, is kept entirely outside training.

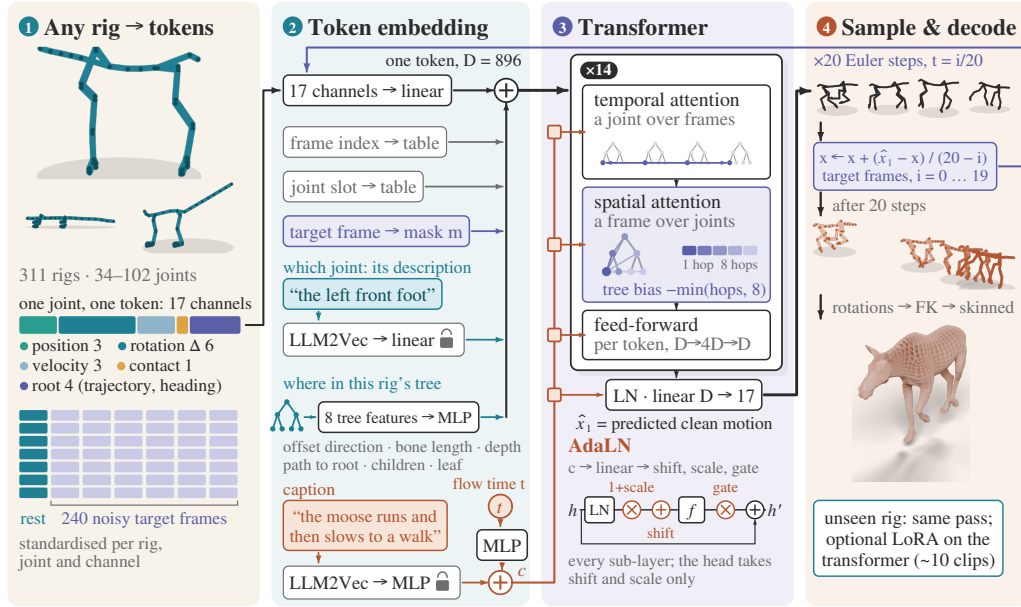


Figure 1: TOPX overview. A rig enters as data: each joint of each frame is one token of 17 standardised channels, and the sequence is the rig’s clean rest pose followed by 240 noise-mixed target frames. A token is the sum of six terms (band 2). The caption and the flow time form c , which sets the shift, scale and gate of every sub-layer (AdaLN) and is the caption’s only route into the network. Fourteen blocks alternate temporal and spatial attention, the latter biased by tree distance; the head predicts the clean motion $\hat{x}_1$, and 20 Euler steps with caption guidance $s=2$ update the target frames only; rotations are then forward-kinematised on the rig and skinned. An unseen rig takes the same pass, optionally with a ten-clip LoRA. All glyphs are the model’s own output on a moose run-to-walk clip.

et al., 2023) degrades as morphology diverges, and a model per character multiplies data and compute by the size of the cast; the motion has to be generated for the target body, and it has to come out as rotations the rig can play.

Our position. A generative model should treat the skeleton as *input*, not as architecture: told the kinematic tree, the rest pose and what each joint *is*, the network needs no per-skeleton parameter, and adding a character becomes a data question. The interface is due to AnyTop (Gat et al., 2025), which prepends the rest pose and embeds joint names to generate, without text, for the Truebones zoo. Text-conditioned generation across skeletons has since been shown by OmniZoo (Chen et al., 2025a) on 140 species, by G-MDM (Lee et al., 2025) on Truebones with joint rotations but no root translation, and by SAMoR (Zhang et al., 2026), which pools any skeleton into eight part tokens over joint positions and recovers rotations by inverse kinematics. Building on skeleton-conditioned generation, we study joint-space rotation and root-trajectory prediction across 311 production rigs at 0.3B parameters, with controlled representation experiments and an external-rig adaptation study; the output is what a rig plays, per-joint rotations and a world-space root trajectory, with no inverse-kinematics fit.

Contributions.

1. **KTJD-17, a per-joint representation with per-cell normalisation** (§3.1). Every joint carries the same 17 channels—position, rotation as a 6D delta from the rest pose, velocity, contact, plus a root-only world trajectory and heading—and every (rig, joint, channel) cell is standardised by its own statistics. Any rigged animation converts to it from its kinematic tree, rest pose and motion, with no template, retargeting, tokenizer or inverse kinematics. Standardising per cell instead of per rig raises R@1 by 2.0–2.1 points and lowers the FID by 22–30% at each of three matched

epochs; against the 13-channel AnyTop layout derived from the same clips, the layout with its world-coordinate loss terms keeps leaf-joint articulation, foot contact and root travel at equal retrieval (§4.5).

2. **In-context skeleton conditioning** (§3.2). The rest pose is prepended as one clean frame, natural-language descriptions of the joints (“left front lower leg”, “tail segment 3”) are added to their tokens, and spatial attention is biased by tree distance. The network holds no per-skeleton parameter: joints occupy positions in a learned table of 160 slots in the order the rig lists them, and what a slot *is* comes from its description, so a new rig is a new set of inputs.
3. **A calibrated grouped flow-matching objective and a recipe that trains at 0.3B** (§3.3). Channel groups are weighted from measured group energies so that the root, contact and trajectory channels of a large rig are not drowned by its many body rotations; a mechanism check certifies the calibration before training; query–key normalisation makes the 0.3B model trainable; and retrieval and FID improve monotonically from 36M to 303M parameters at a matched budget.
4. **An extension study** (§3.5, §4.3). A rig outside the library is served by the same forward pass given its rest pose, joint descriptions and per-cell statistics, and a per-rig low-rank adapter trained on eight to twenty-six clips brings its rotation and position channels into agreement on bodies far from the training distribution (a winged dragon, an eight-legged spider); an adapter shared across species is a negative result.

Results. One 303M-parameter TOPX trained on all 311 rigs is scored by a skeleton-aware text–motion evaluator on 3,899 held-out clips of those rigs: R@1 0.922 against 0.964 for the real clips in the same pools, FID 0.005 (§4.2); the 36M pilot reaches 0.897. On Truebones rigs never seen in training, a per-rig adapter on ten to fifteen clips lowers the FK–pose gap (the disagreement between the rotation and position channels of the prediction) from 1.215 to 0.146 bone lengths on a 95-joint dragon, while the adapted outputs move less than the real clips.

2 RELATED WORK

Text-to-motion generation. Human text-to-motion has moved from VAEs on HumanML3D (Guo et al., 2022) to diffusion in pose space (Tevet et al., 2023; Zhang et al., 2024) or a latent (Chen et al., 2023), distilled for real time (Dai et al., 2024), and to token models over a VQ-VAE (Zhang et al., 2023; Guo et al., 2024), scaled by larger corpora (Lin et al., 2023; Fan et al., 2025) and skeleton-aware latent attention (Hong et al., 2025). All fix one kinematic tree; TOPX keeps the pose-space diffusion and text conditioning and removes the fixed skeleton.

Motion across skeleton topologies. Cross-skeleton motion has mostly meant *retargeting* between two known rigs (Villegas et al., 2018; Aberman et al., 2020) or skeleton-agnostic embeddings such as SAME (Lee et al., 2023). For generation on arbitrary topologies, SinMDM (Raab et al., 2024) fits one motion at a time; AnyTop (Gat et al., 2025) prepends the rest pose as a first frame, adds T5 embeddings of the joint names to every joint token and biases attention by graph distance to generate, without text, for the Truebones zoo (Truebones Motions Animation Studios, 2022); UniMoGen (Khani et al., 2025) is skeleton-agnostic and controlled by style and trajectory. Text-conditioned animal motion has been approached from a fixed 35-joint SMAL template by Omni-MotionGPT (Yang et al., 2024), from a shared human–animal topology by X-MoGen (Wang et al., 2025a) (115 species), across topologies by OmniZoo (Chen et al., 2025a) (a topology-aware skeleton embedding and 140 species), on a unified 30-joint skeleton by AniMo (Wang et al., 2025b), whose AniMo4D captions our corpus carries, on re-captioned Truebones data by G-MDM (Lee et al., 2025), which generates Euler joint rotations without root translation, and, concurrently with this work, by SAMoR (Zhang et al., 2026), which pools any skeleton into eight part tokens over joint positions and velocities and recovers rotations by inverse kinematics. Each compares against baselines adapted or retrained on its own data (SAMoR for cross-topology reconstruction, with published numbers on HumanML3D). We build on AnyTop’s skeleton prompt and differ in what is generated and at what scale: the model is text-conditioned, it denoises a per-joint representation that any rig is converted to without a template, retargeting or inverse kinematics, it embeds richer natural-language *descriptions* of the joints with LLM2Vec where AnyTop embeds cleaned joint names with T5, and every (rig, joint, channel) cell is normalised with its own statistics so that hundreds of rigs of very different proportions share one network.

Objective, encoders, adaptation and evaluation. TOPX is a DiT-style denoiser (Peebles & Xie, 2023) trained with flow matching (Lipman et al., 2023; Liu et al., 2023), classifier-free guidance on the text stream (Ho & Salimans, 2022) and query-key normalisation (Dehghani et al., 2023); its grouped per-channel-family objective follows Kimodo (Rempe et al., 2026). Captions and joint descriptions are embedded with LLM2Vec (BehnamGhader et al., 2024) rather than CLIP (Radford et al., 2021) or T5 (Raffel et al., 2020). New rigs are handled zero-shot through the skeleton prompt, in the spirit of in-context conditioning for image diffusion (Wang et al., 2023), and with a per-rig LoRA (Hu et al., 2022) when they lie far from the library. Retrieval metrics follow the contrastive text-motion evaluators of Guo et al. (2022) and TMR (Petrovich et al., 2023), which are tied to one skeleton; ours is skeleton-aware (§3.6).

3 METHOD

A rig $\mathcal{R} = (\mathcal{T}, \mathbf{p}^{\text{rest}}, \mathcal{D})$ is a kinematic tree $\mathcal{T}$ over J joints, a rest pose $\mathbf{p}^{\text{rest}} \in \mathbb{R}^{J \times 3}$ and a natural-language description $\mathcal{D} = \{d_j\}_{j=1}^J$ of its joints. A motion is a tensor $\mathbf{x} \in \mathbb{R}^{T \times J \times 17}$ in the KTJD-17 representation of Section 3.1. Given $\mathcal{R}$ and a caption c , TOPX samples $\mathbf{x} \sim p_\theta(\mathbf{x} \mid \mathcal{R}, c)$ with one transformer shared by every rig (Section 3.2), trained by a calibrated grouped flow-matching objective (Section 3.3) and sampled in 20 Euler steps (Section 3.4). A rig outside the training library is served by the same forward pass and, when a few clips are available, by a per-rig LoRA (Section 3.5); all retrieval metrics come from a skeleton-aware evaluator (Section 3.6).

3.1 KTJD-17: A PER-JOINT REPRESENTATION ANY RIG CAN BE CONVERTED TO

Channels. Every joint carries the same 17 channels (Table 3 in App. A). Positions are expressed relative to the root’s own ground-plane track, with height absolute, so that the body’s pose and its travel are separated the way animators separate them. Rotations are continuous 6D (Zhou et al., 2019) *deltas* from the joint’s rest orientation in the global frame: “no motion” is the identity rotation, $[1, 0, 0, 0, 1, 0]$ before normalisation, on every rig, which gives the rest frame one meaning across rigs when it serves as the skeleton prompt (Section 3.2). Velocities are supervision only and never integrated at decode time; the root is recovered directly from the trajectory channels and the per-clip origin. Everything is expressed in one canonical frame (+Y up, rest forward +Z) obtained by a rigid per-rig recanonicalisation to the foot support plane, after which forward kinematics is unchanged to numerical precision (App. A). Building KTJD-17 for a new rig needs only its kinematic tree, rest pose and animation: there is no template skeleton, no retargeting and no inverse-kinematics fitting, and the same channel definition and codec convert every Truebones rig used in Section 3.5.

Per-cell normalisation. Rigs differ in size, proportion and convention by orders of magnitude, so every (rig, joint, channel) cell is standardised with its own mean and standard deviation (σ floored at 0.05), and cells that are constant across a rig’s library (9,123 never-firing contact flags, for instance) are removed from supervision. The statistics are a property of the rig, computed from its motion library and supplied with it at deployment, the footing on which motion-transfer tools require example motions of the target character (Chen et al., 2025b).

Corpus. The library holds 311 rigs, 77,894 captioned clips and 6.18M frames at 30 fps (train 73,995 / validation 3,899), deformation bones only, 34–102 joints per rig (median 64; 108 distinct kinematic trees, 123 distinct ordered parent arrays). The Truebones zoo (66 rigs, 986 clips) is never trained on by the backbone; App. A gives the corpus statistics and the conversion.

3.2 IN-CONTEXT SKELETON CONDITIONING

Layout. The network input is $[\mathbf{x}^{\text{demo}} \mid \mathbf{x}_t^{\text{tgt}}]$ along time: one clean frame holding the rig’s analytic rest pose, never noised, followed by the $T=240$ target frames that carry the flow interpolant and are the only frames denoised and scored. Under the rest-delta parameterisation the rest frame’s rotation channels are the identity on every rig, so it conveys proportions, joint count and graph rather than motion (after per-cell normalisation the prompt also carries the rig’s statistics); this is AnyTop’s rest-pose first frame (Gat et al., 2025), which we keep.

Joint semantics. Each joint description d_j is embedded once with LLM2Vec (BehnamGhader et al., 2024) (4096-d), projected and *added* to every token of joint j ; descriptions rather than bone names, because “front left upper leg” is the same concept on a horse and a lizard whereas `L_Femur_02` and `b_hindleg_l_1` are not, and a new rig’s naming scheme need never have been seen.

Structure and graph bias. An 8-d structural vector per joint (rest-offset direction, bone length, tree depth, root-path length, child count, leaf flag) enters through a zero-initialised MLP, and spatial attention receives a fixed bias from the geodesic distance on the tree plus two learned tables indexed by the hops to the lowest common ancestor (App. B).

Block. Each of the 14 blocks applies temporal attention (tokens are frames, batched over joints), spatial attention (tokens are joints, batched over frames, with the graph bias) and a $4\times$ MLP, each modulated by AdaLN (Peebles & Xie, 2023) from $\mathbf{c} = \text{MLP}(t) + \text{MLP}(\mathbf{e}_c)$, with $\mathbf{e}_c$ the pooled LLM2Vec caption embedding, so text reaches the model only through modulation and the token stream stays kinematic. Width 896, 14 heads, 302.9M parameters.

Stability and guidance. Queries and keys are RMS-normalised per head before the dot product (Dehghani et al., 2023): in our runs every 0.3B attempt without it diverged with attention logits growing from $\sim 1.4\text{k}$ to 23k, and with it the eight probes taken through epoch 199 of the 303M run peak at 28.1. The caption is dropped with probability 0.1 for classifier-free guidance; the rest-pose prompt is never dropped.

3.3 CALIBRATED GROUPED FLOW-MATCHING OBJECTIVE

Flow matching with $\mathbf{x}$ -prediction. We train on the straight path $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$ with $\mathbf{x}_0 \sim \mathcal{N}(0, I)$ and $t \sim \mathcal{U}(0, 1)$ (Lipman et al., 2023; Liu et al., 2023); the network predicts the clean sample $\hat{\mathbf{x}}_1$, and the velocity-space weight of Li & He (2025) is applied,

$$\ell(\mathbf{x}_t, t) = \frac{w(t)}{\mathbb{E}[w]} \text{Huber}_\delta(\hat{\mathbf{x}}_1 - \mathbf{x}_1), \quad w(t) = \frac{1}{\max(1 - t, \sigma_{\min})^2}, \quad (1)$$

with $\sigma_{\min} = 0.2$, $\delta = 10$ (below the measured 99.99th percentile of the normalised targets, 11.45; the knee acts on residuals, and at initialisation it binds on 0.015% of supervised cells, recorded in the calibration artifact) and a uniform t sampler (App. D).

Grouped loss. A rig with 102 joints has three times the body cells of one with 34 and the same single root row, so a plain mean over cells would drown the root trajectory, heading and contact channels—the ones that decide whether a walk is a walk—in body rotations on large rigs. We partition the channels into nine groups g (root and body $\times$ position, rotation, velocity; contact; root track; heading) and use

$$\mathcal{L} = \sum_g \gamma_g \sqrt{\frac{N_g}{N}} \bar{\ell}_g, \quad (2)$$

where $\bar{\ell}_g$ is the masked mean of Eq. equation 1 over the N_g valid cells of group g and $N = \sum_g N_g$; the grouping by channel family follows Kimodo (Rempe et al., 2026). Under this weighting the explicit N_g dependence of the output-gradient energy of group g cancels: it is proportional to $\gamma_g^2 Q_g$, where Q_g is the group’s mean weighted squared residual, so a group’s weight in $\partial\mathcal{L}/\partial\hat{\mathbf{x}}_1$ no longer grows with how many joints the rig has (App. C). The shares realised during training still follow the evolving residuals; the weighting removes the joint-count bias, it does not fix the shares.

Calibration. The γ_g are not hand-tuned. A target share profile over channel families is pre-registered, the group energies E_g are measured over one pass of the training set under exactly the training crops and masks, and the weights are solved as $\gamma = \sqrt{\text{share}/E}$, with the rotation family anchored at $\gamma=10$ and the others solved relative to it. A *mechanism check* then takes 30 optimiser-free steps of a small arm model and verifies that every family’s measured share of $\|\partial\mathcal{L}/\partial\hat{\mathbf{x}}_1\|^2$, accumulated over those steps, matches the $\gamma_g^2 Q_g$ prediction within 25%. It certifies the wiring of the weights, not that the pre-registered profile is realised during training: the realised share moves with

the error energies, which evolve. The trainer refuses to start unless the artifact matches the objective it is run with (t sampler, $\sigma_{\min}$, Huber knee, masks, batch size; App. C).

Auxiliary terms. A forward-kinematics consistency term between predicted positions and positions forward-kinematised from predicted rotations, a velocity term on world joint displacements, a foot-lock term gated by the contact channel, and an acceleration-matching term are added in the same frame (App. C).

Optimisation. AdamW, learning rate 1.5×10^{-4} at global batch 128 (the 36M pilot: 2×10^{-4}), 4k warm-up steps, cosine decay to a 10^{-2} floor over 40 epochs, bf16, gradient clipping at 1.0, and rejection of any step whose pre-clipping gradient norm exceeds 200; 8 H200 GPUs, 578 steps per epoch, 290 epochs (best validation loss at 289).

3.4 SAMPLING

Samples follow the straight path with Euler steps $\mathbf{x} \leftarrow \mathbf{x} + (\hat{\mathbf{x}}_1 - \mathbf{x})/(S - i)$, $S=20$; invalid cells are projected back to zero after every step, and guidance is applied on the text stream only, $\hat{\mathbf{x}}_1 = \hat{\mathbf{x}}_1^{\emptyset} + s(\hat{\mathbf{x}}_1^c - \hat{\mathbf{x}}_1^{\emptyset})$ with $s=2$. Twenty steps is the operating point used throughout; we did not tune it further.

3.5 EXTENDING THE MODEL TO A RIG IT HAS NEVER SEEN

Zero-training arm. An unseen rig is served by the frozen backbone as soon as its rest pose, joint descriptions and per-cell statistics (from its own clips) are supplied; on the Truebones rigs this yields motion with far more jitter than the real clips, since wing chains and eight-legged gaits are absent from the training prior.

Per-rig LoRA. When roughly ten clips of the new rig exist, we attach rank-64 or rank-128 adapters (Hu et al., 2022) (scale 1) to the attention projections, the MLPs and the conditioning paths (joint-semantics projection, caption MLP, structural MLP, AdaLN), leaving the input and output projections and the timestep MLP frozen; the adapters hold 26.8M (r64) or 53.6M (r128) parameters and are folded back into the weights at export, so every consumer loads a plain checkpoint. The new rig’s data view is built by the corpus script (joint pruning to the deformation-bone convention, corpus-style captions, a split by source animation) and the adapter is trained with the backbone recipe for 300–900 epochs (App. D); a single adapter shared by a group of rigs is reported as a negative result.

3.6 SKELETON-AWARE EVALUATION

Evaluator. Retrieval metrics require a text–motion embedding, and existing ones are tied to one skeleton. We train two towers into a shared 512-d space: a frozen DistilBERT text tower with a trainable pooling head, and a graph-aware motion tower (6 graph layers, 4 temporal layers) over the KTJD-17 channels without contact, with a symmetric InfoNCE objective whose false-negative mask ties together clips of the same motion, the same source animation and the same caption. The evaluator sees only real data, never generated samples, and shares no weights with the generator.

Protocol. Text→motion R@1/2/3 in pools of 64, matching score and FID in the evaluator space, each next to the evaluator’s score on the real clips of the same pools (with shuffled captions the evaluator scores R@1 0.035 in pools of 32, chance 0.031); generation is fixed at 20 steps, $s=2$ and one seed and sharded across GPUs under a hashed generation plan with provenance checks. Rigs outside the library, for which no evaluator exists, are scored geometrically (§4.1).

4 EXPERIMENTS

We ask four questions. (1) How well does one set of weights serve hundreds of heterogeneous skeletons on text-conditioned generation (§4.2)? (2) How far does the model extend to a rig it has never seen, with no training and with about ten clips (§4.3)? (3) How does it relate to the released

arbitrary-topology generator and to prior work in capability (§4.4)? (4) Which ingredients matter (§4.5)?

4.1 SETUP

Library and splits. The training library is the 311-rig animal corpus of §3.1 (77,894 clips, 6.18M frames at 30 fps, 34–102 joints). Validation holds 3,899 clips over 311 rigs (5% of every rig’s clips, drawn by a one seeded permutation per rig; App. K). For rigs outside the library we use the Truebones zoo (Truebones Motions Animation Studios, 2022): 986 clips over 66 rigs after conversion to KTJD-17 with the same pipeline (802 of them carry a caption). Each Truebones rig is split 80/20 into train and validation clips by source animation, so that a clip and its cut neighbours never straddle the split.

Models. The main model has 303M parameters (14 blocks, width 896, 14 heads); a 36M pilot (8 blocks, width 384, 6 heads) shares every other setting and is used for ablations. Both train at global batch 128 with the calibrated objective, the main model for 290 epochs at 1.5×10^{-4} and the pilot for 400 at 2×10^{-4} , with a 4,000-step warm-up and a half-cosine decay over 40 epochs to 1% of the peak.

Protocol. Generation uses 20 Euler steps and text guidance $s=2$ everywhere (§3.4); each clip is generated at the length of the real clip, capped at 240 frames, with one seed. Retrieval metrics follow §3.6: text-to-motion R-precision at $K=1, 2, 3$ in pools of 64 formed by chunking the dataset order (a retrieved motion is correct if it is the caption’s clip, another export of the same source motion, or a clip with an identical caption), the FID on the evaluator’s ℓ_2 -normalised embeddings, and the text–motion matching score, each next to the evaluator’s score on the real clips of the same pools (a reference, not a bound). Because a skeleton-aware evaluator exists only for the library rigs, the Truebones experiments are scored geometrically, in world space, on the deployable output (the forward kinematics of the predicted rotations): the articulation ratio (mean joint speed relative to the root, divided by the real clips’ value), the jitter ratio (mean second temporal difference, likewise normalised) and the root speed ratio. Ratios are taken against *all* real clips of the rig at a common 20 Hz, so that AnyTop samples and ours are measured identically. The *FK–pose gap* is the mean distance, in the rig’s mean bone length, between the joint positions obtained by forward kinematics of the predicted rotations and the directly predicted ones: a consistency of the two channels, not a distance to the target, read from the training-time validation pass (App. C). Library clips are generated at the real clip’s length (at most 240 frames) from the epoch-289 checkpoint with seed 42; the Truebones arms of §4.3 use the epoch-214 snapshot that initialised the adapters, seed 7.

4.2 ONE MODEL FOR 311 SKELETONS

Table 1 reports text-to-motion retrieval on the validation set. All 311 rigs appear in training, so this measures the generation of held-out clips on known rigs; rigs outside the library are the subject of §4.3. The 303M model reaches R@1 0.922 against 0.964 for the real clips scored by the same evaluator in the same pools, a text–motion matching score of 0.764 against 0.782, and an FID of 0.005; the 36M pilot, trained on the same data with the same recipe, reaches R@1 0.897 and FID 0.008. The pools follow the dataset order, so a pool holds clips of one rig or of neighbouring rigs; R-precision uses the 3,840 clips of the 60 complete pools, the matching score and the FID all 3,899. The 95% bootstrap interval over pools is [0.907, 0.936] for the 303M R@1 and [0.955, 0.971] for the real clips (App. H).

What the score responds to. Five controls (App. H) locate the 303M model’s R@1 of 0.922. Re-deriving its joint positions from its own rotations leaves 0.923; permuting its frames leaves 0.805. Two degenerate sequences built from real data set the floor: the target clip’s own first pose, held for its duration, scores 0.346, and the rig’s rest pose 0.109, against a chance level of 0.018. Giving every clip in a pool another clip’s caption leaves 0.005. A static pose of the right species is therefore already worth 0.346 under this protocol, which is the level any retrieval number here should be read against.

Table 1: Text-to-motion generation on the validation set (3,899 clips over 311 rigs; 60 pools of 64, 20 steps, $s=2$, one seed; App. K gives the other pool sizes). The reference row scores the real clips with the same evaluator in the same pools. R-precision is text→motion; Match is the mean text→motion cosine; FID is on the evaluator’s normalised embeddings (lower is better).

	Params	R@1↑	R@2↑	R@3↑	FID↓	Matching↑
Real clips (reference)	—	0.964	0.998	0.999	0	0.782
TOPX (pilot)	36M	0.897	0.966	0.982	0.008	0.746
TOPX	303M	0.922	0.982	0.992	0.005	0.764

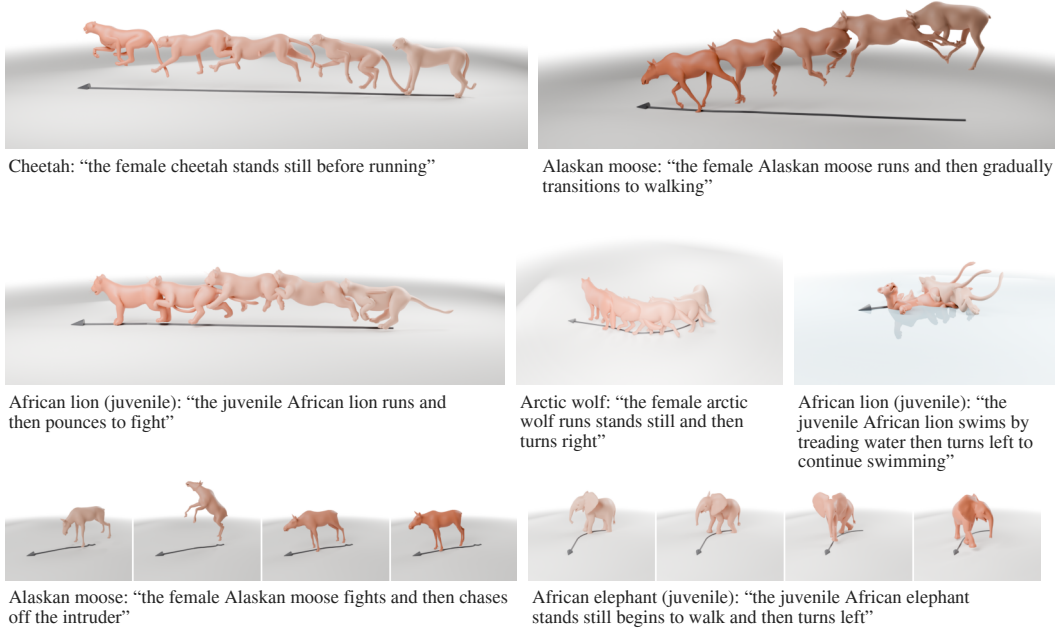


Figure 2: Seven generated clips of five library rigs, rendered on their production meshes. For traveling actions, four to five poses at equal distances along the root path, shaded from light (first) to dark (last), with the root trajectory and its direction on the ground; for actions on the spot, four frames at equal times from one fixed camera. The ground plane sits at the level the feet occupy most often; the water plane in the swimming clip is placed just below the root. Captions are the clips’ own.

4.3 EXTENDING THE MODEL TO RIGS IT HAS NEVER SEEN

Zero training and per-rig LoRA. We take four Truebones rigs that differ from the library in body plan and joint count – Buffalo (30 joints), Gazelle (29), Dragon (95 joints, a flying rig) and Spider (68) – and compare three arms on the same held-out clips of each rig: (a) the library model with the new rig’s per-cell statistics, computed from all of its clips including the evaluation clips, as its only rig-specific input (*zero training*); (b) a per-rig low-rank adapter trained on the rig’s 8–26 training clips (*LoRA*, §3.5); (c) a 36M model of the same recipe trained on all 802 captioned Truebones clips, evaluation clips included, as a seen-clip reference rather than a competing method. Table 2 gives the FK-pose gap on the held-out clips before and after adaptation next to the articulation and jitter of the sampled output: eight to twenty-six clips bring the gap from 0.25–1.2 bone lengths to 0.08–0.15, most on the two rigs farthest from the library, and the seen-clip reference sits at 0.06–0.13.

Table 2: Extending TOPX (303M) to four Truebones rigs, as a diagnostic given the rig’s statistics: the per-cell statistics and masks of each rig are computed from all of its clips, the evaluation clips included, and the adapter checkpoint is the one with the lowest validation loss on those clips (a support-only protocol is reported in the final version). *zero*: library model with the rig’s statistics only (gap read after at most 15 adapter warm-up steps); *LoRA*: per-rig adapter at its best validation loss; *TB-only*: a 36M model trained on all Truebones clips including these (seen-clip reference; its gap is read the same way as the zero column). Gap in bone lengths; articulation and jitter of the sampled FK output relative to all real clips of the rig (1 = real). The same study on the 36M pilot backbone, and a single adapter shared by eight flying rigs, are in App. E.

Rig	Train / val	FK-pose gap (bl)↓			Articulation ratio			Jitter ratio		
		zero	LoRA	TB-only	zero	LoRA	TB-only	zero	LoRA	TB-only
Buffalo (30 j.)	15 / 5	0.323	0.132	0.072	1.45	0.30	0.67	3.32	0.33	1.02
Gazelle (29 j.)	8 / 2	0.255	0.084	0.062	5.39	0.77	1.22	13.2	0.86	1.92
Dragon (95 j.)	10 / 3	1.215	0.146	0.134	1.02	0.49	1.23	2.53	0.52	1.12
Spider (68 j.)	26 / 6	1.029	0.130	0.088	2.87	0.24	0.35	6.49	0.20	0.37

4.4 RELATION TO PRIOR WORK

Capabilities. Table 6 (App. F) places TOPX among the generators that serve more than one skeleton, by what each returns and what a rig needs to play it. TOPX returns per-joint rotations and a world-space root trajectory the rig plays as generated, and measures that geometry against ground truth (Table 11).

AnyTop as an in-distribution reference. The released AnyTop (Gat et al., 2025) weights trained on the whole Truebones zoo, including the four rigs of Table 2; its per-rig geometry (App. G) is therefore a reference for what a model trained on these rigs produces.

4.5 ABLATIONS

The ablations share the pilot’s recipe and checkpoint epoch; App. I collects them, with the notes. Removing the acceleration-matching term leaves the FK-pose gap, retrieval and FID unchanged at epoch 200 (Table 9). Normalising KTJD-17 by a per-rig scale alone instead of per-cell mean and standard deviation costs about two points of R@1, and 22–30% of the FID, at every matched epoch (Table 10); replacing its channel layout by the 13-channel AnyTop layout derived from the same clips moves R@1 by under a point at every matched epoch (Table 12), while the 17-channel layout keeps the leaf-joint articulation, foot contact and root travel that the 13-channel layout loses (Table 11), and keeps its two output channels consistent: re-encoding a sample through the forward kinematics of its own rotations costs 0.005 of R@1 on KTJD-17 against 0.035 on the 13-channel layout (App. H). Running KTJD-17 under the 13-channel arm’s own objective and partition puts the two layouts 0.007 of R@1 apart, so the retrieval numbers are measuring the objective rather than the layout (App. I). At a matched budget of 200 epochs the 36M, 116M and 303M models reach R@1 0.878, 0.900 and 0.914 (Table 13).

5 CONCLUSION

A text-to-motion model can treat the skeleton as an input rather than as part of the architecture: one 303M-parameter model trained on 311 animal rigs generates text-conditioned motion for all of them (R@1 0.922 against 0.964 for the real clips), returning rotations and a root trajectory the rig plays directly, and a low-rank adapter on eight to twenty-six clips extends it to rigs outside the library.

Limitations. The library is animal-only; retrieval comes from an evaluator we train, hence the real-motion reference beside every number and the controls of App. H; a rig’s statistics come from its own clips, so a new rig needs clips or supplied ones; and the extension study is four case studies with two to six validation clips each.

REPRODUCIBILITY STATEMENT

Section 3.1 and Appendix A specify the representation channel by channel, the conversion from a kinematic tree, rest pose and motion, and the per-cell normalisation; Appendix C gives the calibration protocol, the target share profile and the mechanism check that certifies an artifact; Appendix D lists every adapter hyper-parameter; Appendix G gives the AnyTop sampling protocol and the geometric protocol, including the frame-rate handling. All evaluation runs are seeded, and the sharded generation used for retrieval metrics records the checkpoint, sampler and source hashes so that a table can be re-scored from the stored samples.

AI USE STATEMENT

We used generative AI tools throughout this work. The code for data conversion, training, evaluation and rendering was drafted with an LLM-based coding assistant and independently reviewed by a second LLM reviewer before it was run; every script was tested, and every number in this paper was checked against the run logs and artifacts by the authors. LLM assistance was also used for experiment orchestration, for auditing the data, for figure scripts, literature search and the drafting and editing of the text. The joint descriptions were produced with an LLM from the joint names; the clip captions are the AniMo4D annotations (Wang et al., 2025b). The research questions, the design decisions and the interpretation of the results are the authors’, and we take responsibility for the final content, including text, claims and artifacts produced with the aid of generative AI.

ETHICS STATEMENT

The work involves no human subjects and no personal data. The training library follows the clip inventory and captions of AniMo4D (Wang et al., 2025b), and the evaluation rigs come from the publicly distributed Truebones zoo. The paper specifies the conversion channel by channel, so the representation can be rebuilt from the same sources. Generated animal motion for games and film carries no risk beyond the generic misuse of animation tools.

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Table 3: KTJD-17 channel layout per joint and frame. Channels 13–16 exist on the root row only; every (rig, joint, channel) cell is normalised with its own statistics and exactly-constant cells are excluded from supervision.

slice	content	frame	supervised on
0:3	position (XZ relative to the root track, Y absolute)	world	all joints
3:9	rotation, continuous 6D delta from the rest orientation	global	free-DOF joints
9:12	world velocity (supervision only, never integrated)	world	all joints
12	contact flag	—	joints with contact
13:15	root trajectory XZ	world	root only
15:17	heading ($\cos \phi, \sin \phi$)	world	root only, valid frames

A KTJD-17 DETAILS

Validity. Not every cell exists on every rig or frame. Frame and joint masks handle padding; rotation is supervised only on joints with free degrees of freedom, contact only on joints that can touch the ground, heading only on frames where it is defined. The per-rig cell mask multiplies the interpolant, the loss and the per-step projection at sampling time, so an invalid cell is exactly zero everywhere rather than “small”. Per-frame heading validity is the one part of the mask that training has and sampling does not: it gates the target’s heading channels during training, while at generation the target’s heading is what is being generated, so the sampler carries the demonstration’s flag only and the target heading is produced unconditionally.

Why standardise rather than scale. We found this to be decisive. Under the scale-only normalisation of the source protocol a large part of the objective is already explained by a constant pose, and the network settled on a frozen pose while the root travelled. On a 671-window probe of an earlier build, a constant prediction recovers 75% of the scale-only objective; centring each cell on its mean brings that to 41.6%, and centring and dividing by the standard deviation, the normalisation used here, brings it lower still. Table 10 measures the consequence directly on the corpus used here: at every matched epoch the standardised representation is about two points of R@1 ahead, with the FID 22–30% lower.

Conversion. A clip enters as a kinematic tree, a rest pose and per-frame local rotations and root translation. It is first resampled to 30 fps (linear interpolation of the root translation, spherical interpolation of the local rotations, then forward kinematics) and expressed in a right-handed canonical basis. Global joint rotations follow from forward kinematics; the *heading* is the ground-plane direction between a designated pair of joints of the rig, stored as the unit vector (f_z, f_x) and marked invalid (channels set to zero, excluded from supervision) on frames where that projected direction has length below 10^{-6} . The *root track* is the root joint’s own XZ trajectory, stored unsmoothed on the root row; joint positions store XZ relative to it and Y as absolute height, so every window can be re-based to its own first frame without touching the pose. Rotations are the global rotation composed with the inverse rest rotation of the same joint, encoded as the first two matrix columns (continuous 6-D), so a joint at rest is the identity on every rig. Velocities are one-step finite differences of world positions, per second, supervised but never integrated. A contact flag marks the frames on which a joint that can touch the ground is within 0.3 of the lowest point its rig’s contact joints reach in the clip and whose squared displacement to the next frame is at most 0.002 (canonical units; both tests read the following frame, so the flag at frame t describes the step from t to $t+1$). The Truebones corpus was converted by an earlier build of the same codec, which smoothed the root track with a 1 Hz zero-phase filter, took the heading from the carrier joint’s rotation with a 0.05 validity floor, and set contact from normalised height and speed (0.05 and 0.25); the main-body pruning of Appendix D precedes it. Both builds write the same 17 channels with the same meanings, which is what lets one model serve a rig from either corpus, and every Truebones number here is measured within that corpus.

Per-cell statistics. For every rig, joint and channel the mean and standard deviation are taken over all frames of all of the rig’s clips (heading channels over valid frames only), and a cell is served as $(x - \mu)/(\sigma + 10^{-6})$ with σ raised to at least 0.05. A cell whose value never changes on a rig

(contact of a joint that never touches the ground, a fixed degree of freedom) is exactly constant; it is normalised to zero, excluded from supervision and held at its constant when decoding. On the library this concerns 9,185 of the 256,512 cells that exist on the 311 rigs, 9,123 of them contact flags. A rig outside the library is served the same way from statistics computed on its own clips; how far a rig can be served without them is the subject of §4.3.

Corpus statistics. The library holds 311 rigs and 77,894 clips (73,995 training, 3,899 validation), 6.18M frames at 30 fps, or 57 hours. Clips are short: median 62 frames, mean 79, maximum 299, and 2.1% exceed the 240-frame training window. Rigs have between 64 and 946 clips (median 236). Joint counts run from 34 to 102 (median 64): 61 rigs in the thirties, 48 in the forties, 14 in the fifties, 63 in the sixties, 48 in the seventies, 27 in the eighties, 47 in the nineties and 3 above a hundred. The clips and their captions follow the official AniMo4D annotation set of Wang et al. (2025b): 77,894 of its 78,149 entries have a processed clip here and the remaining 255 have none. 53,448 clips carry three captions and 24,443 carry one (three carry more), about ten words each at the median; the primary caption is the one used for training and for evaluation. One caption of one clip was dropped as corrupt (a 12,296-character repetition); its two siblings were kept. The rigs are our own re-export with deformation bones only, the natural-language joint descriptions are ours (written from the joint names, §3.2), and the conversion to KTJD-17 is new. A source audit of an earlier build of this corpus identified 161 clips whose joint speeds lay in the extreme tail of their rig, which trace to defects of the source animations rather than to the conversion. Of those 161, 81 are KTJD-17 library clips and all 81 are present in the corpus used here (78 training, 3 validation): they are not removed, the objective’s robust knee (§3.3) having been adopted in their place.

B MODEL DETAILS

Structure and graph bias. An 8-d structural vector per joint (unit rest-offset direction, log bone length, tree depth, root-path length, child count, leaf flag) enters through a zero-initialised MLP. Spatial attention receives a fixed bias $-\min(g_{ij}, 8)$ from the geodesic distance g_{ij} on the tree, plus two learned tables indexed by the number of hops up and down to the lowest common ancestor (clipped at 15), so that “parent”, “sibling” and “tail of the same limb” are distinguishable; padded joints receive -10^4 .

C CALIBRATION OF THE GROUPED OBJECTIVE

Batch of the mechanism check. The grouped loss normalises per batch, so an artifact certifies the objective at the batch it was checked with, and the trainer refuses a mismatch. The two models of Table 1 predate that rule: they continue under an artifact checked at batch 8 while training at 16 per rank, which the run announces and records in its checkpoint. Every arm started after the rule carries a batch-matched artifact.

Auxiliary terms. Four terms are added in the same calibrated frame: a forward-kinematics consistency term between predicted positions and positions forward-kinematised from predicted rotations (weight 0.07, ramped over the first 5k steps); a velocity term matching world joint displacements between consecutive frames over every valid joint (0.01); a foot-lock term, gated by the de-normalised contact channel, that demands zero displacement of a joint flagged in contact (0.01); and an acceleration-matching term on second temporal differences (1.0), which is what removed the residual jitter of early runs.

Groups, families and target shares. The 17 channels form nine groups (root position, root rotation, root velocity, body position, body rotation, body velocity, contact, root track, heading), collected into six families with a prescribed share of the objective: root trajectory (root position and root track) 0.304, heading 0.012, body position 0.304, rotation (body and root) 0.304, velocity (body and root) 0.027 and contact 0.049. Each group’s energy E_g is the mean squared standardised target over its valid cells, measured in one pass over the training cohort at the run’s batch and demonstration condition; the family weight is then the closed form $\gamma_f = \sqrt{s_f / \sum_{g \in f} E_g}$, applied to every group of the family, with the rotation family anchored at 10 and the others scaled relative to it. On

the library this gives 12.2 for the root trajectory, 14.2 for body position, 10 for rotations, 3.0 for velocities, 5.6 for contact and 2.8 for heading.

Mechanism check and artifact. Because the weights act on gradients rather than on targets, the artifact is certified by running 30 optimiser-free steps of a small model under the run’s exact objective (velocity-space loss, $\sigma_{\min}=0.2$, Huber knee 10, uniform t) and requiring the gradient share of every family, accumulated over those steps, to lie within 25% of $\gamma_f^2 \sum_{g \in f} Q_g$, where Q_g is the group’s mean time-weighted, Huber-clipped squared residual. This is a check of the wiring: it says the weights act on the gradient the way the solve assumes, at initialisation. It says nothing about the share realised during training, which moves with the error energies – the artifact itself records that it cannot verify the design target. The artifact records the corpus generation, the normalisation variant, the statistics file, the batch, the demonstration condition and the loss code; a derived training view records its manifest, clip set and cut as well. The trainer refuses to start on any mismatch of what the artifact records, so a calibration cannot be reused across a change of data, batch, normalisation or objective without being re-measured.

D ADAPTER SETTINGS

Data view of a new rig. The data view of a new rig is built by the corpus script with three additions: joints are pruned to the deformation-bone convention of the training library (end markers, cosmetic and helper joints dropped by name or description, never the root, heading carrier, trunk or wing joints, and only leaf-closed subtrees so no forward kinematics is re-run); captions are rewritten to the corpus style “The <species> <action>”; and the train/validation split is made by source animation so that no clip straddles it. Training follows the backbone recipe with the learning rate scaled linearly to the batch, one pass over every clip per epoch, 300–900 epochs and a shortened forward-kinematics warm-up. A single adapter can also be shared by a group of rigs; we report this as a negative result.

Hyper-parameters. Table 4 lists the per-rig settings. The learning rate follows the linear-scaling rule from 10^{-4} at batch 8; the batch divides the rig’s training-clip count so that every clip is seen exactly once per epoch, and is at most 13. Adapters act on the query, key, value and output projections of both attention blocks, the feed-forward layers and the conditioning MLPs (62 linear layers of the 36M backbone, 104 of the 303M one); a 100-step warm-up precedes a half-cosine decay over five sixths of the epochs to 1% of the peak, and the forward-kinematics term ramps over 100 steps. At rank 64 the adapter holds 7.06M trainable parameters on the 36M backbone and 26.8M on the 303M one (16% and 8% of the respective totals); Dragon uses rank 128 (53.6M, 15%) and three times the epochs.

Table 4: Per-rig adapter settings. Clips are the usable Truebones clips after the source audit, split by source animation.

Rig	Joints	Train / val clips	Batch	Learning rate	Rank	Epochs
Buffalo	30	15 / 5	5	6.25×10^{-5}	64	300
Gazelle	29	8 / 2	8	1.0×10^{-4}	64	300
Dragon	95	10 / 3	5	6.25×10^{-5}	128	900
Spider	68	26 / 6	13	1.63×10^{-4}	64	300

E ADDITIONAL EXTENSION RESULTS

Smaller backbones. Repeating the study with the 36M pilot as the backbone gives the same picture: the zero-training arm is less jittery than the 303M zero-training arm on every rig (jitter ratio 1.92 / 6.24 / 0.74 / 4.71 vs. 3.32 / 13.2 / 2.53 / 6.49) with articulation 1.06 / 2.48 / 0.21 / 1.93, and the LoRA arm is close to the 303M LoRA (0.27 / 0.92 / 0.62 / 0.23). Rank 64 is 16% of the trainable parameters of a pilot and 8% of the main model; these are same-rank, not same-relative-capacity, controls. The Truebones-only 36M model, which has trained on the evaluation clips, reproduces the real clips’ statistics on every rig (jitter ratio 1.02 / 1.92 / 1.12 / 0.37 and articulation 0.67 / 1.22

/ 1.23 / 0.35 against all clips of the rig; 0.95–1.29 and 0.85–1.00 against the clips it was asked to reproduce), unchanged between 30k and 60k steps. Against those matched targets it is the one arm whose jitter sits at the real value; the LoRA arms come out *smoother* than real motion under the all-clip denominator used in Table 2 (jitter ratios 0.2–0.9), and the Truebones-only model does so on Spider alone (0.37).

Sharing an adapter across species does not help. One rank-128 adapter trained on the pooled clips of eight flying Truebones rigs (Bat, Bird, Buzzard, Dragon, Eagle, Giant bee, Parrot, Pteranodon; 78 training clips) is scored on the validation clips of each rig with the geometric protocol of §4.1 (all real clips of the rig as the reference; at the rigs’ native 30 Hz, as no external source is involved). Table 5 compares it with the zero-training arm on seven rigs. The shared adapter removes the zero-training arm’s jitter excess (median 5.5 to 0.99 times real) and settles articulation at 0.46–0.83 of the real clips. On Dragon, the one rig that also has an adapter of its own, the two are close: articulation 0.52 shared against 0.49 species-specific and jitter 0.95 against 0.70, both nearer the real motion’s 1 for the shared adapter, while the species-specific one is nearer on root jitter. Pooling eight bodies therefore buys smoothness on the rigs that have no adapter of their own, and does not replace one where it exists.

Table 5: A shared adapter over eight flying rigs against zero training (303M backbone, FK output relative to all real clips of the rig, 30 Hz; 1 = real). n : validation clips. Pteranodon is omitted: decoding its ground-truth clip for the comparison hit a degenerate 6D vector at its first frame and the renderer refused it.

Rig	n	Articulation ratio		Jitter ratio	
		zero	shared LoRA	zero	shared LoRA
Bat	2	1.79	0.51	5.52	0.57
Bird	3	3.89	0.81	12.2	1.34
Buzzard	2	1.40	0.46	4.18	0.38
Dragon	3	1.14	0.52	4.28	0.95
Eagle	2	3.88	0.83	13.2	0.99
Giant bee	2	1.72	0.69	4.28	1.02
Parrot	3	1.84	0.55	7.14	1.14

F CAPABILITIES OF RELATED SYSTEMS

Table 6: Capabilities of motion generators across skeletons, as described in the papers and released code. *Any topology*: one model serves skeletons of different joint counts and trees. *Text*: language-conditioned. *Rotations*: joint rotations are predicted. *IK-free*: the exported animation needs no inverse-kinematics fit. *Root*: a world-space root trajectory is generated. *Unseen rig*: a rig absent from training is served, and how. *Baselines*: how the paper compares to other methods. n.r. = not reported. SAMoR is its cross-topology variant; its HumanML3D variant has a root and rotation branch. G-MDM omits root translation in its current implementation.

	Any topology	Text	Rotations	IK-free	Root	Unseen rig	Baselines
Human T2M diffusion (Tevet et al., 2023; Zhang et al., 2024)	×	✓	✓	×	✓	×	standard HumanML3D protocol
AniMo (Wang et al., 2025b) (30-joint skeleton)	×	✓	✓	✓	✓	×	retrained human T2M models
OmniMotionGPT (Yang et al., 2024)	×	✓	✓	✓	✓	×	retrained on its data
X-MoGen (Wang et al., 2025a)	×	✓	×	n.r.	n.r.	×	retrained on its data
AnyTop (Gat et al., 2025)	✓	×	✓	×	✓	zero-shot, qualitative	MDM adapted, SinMDM per rig
OmniZoo (Chen et al., 2025a)	✓	✓	✓	n.r.	n.r.	qualitative demos	padded MoMask / MMM / AniMo
G-MDM (Lee et al., 2025)	✓	✓	✓	✓	×	qualitative	per-object MDMs, retargeting
SAMoR (Zhang et al., 2026) (cross-topology)	✓	✓	×	×	×	held-out characters	T2M-GPT / MoGenTS retrained (reconstruction)
TOPX (ours)	✓	✓	✓	✓	✓	given its statistics, + adapter	released AnyTop as reference

G ANYTOP AS AN IN-DISTRIBUTION REFERENCE

In-distribution references, not a comparison. AnyTop (Gat et al., 2025) is the released arbitrary-topology generator; it is unconditional and was trained on the whole Truebones zoo, including the four rigs above, so on them it is an in-distribution model. Of our own arms only the unadapted

Table 7: Per-rig descriptive kinematics on the Truebones zoo, on the deployable output, relative to all real clips of the rig (1 = real). AnyTop: released weights, 8 samples per rig, unconditional, *trained on these rigs* (an in-distribution reference, not a competitor). Ours: 303M model, held-out clips, caption-conditioned; *zero* = statistics only, *LoRA* = per-rig adapter (FK-pose gap as in Table 2). Spider’s real root is static, so its root-speed ratio is undefined. n = number of sequences.

Rig	Source (n)	Jitter	Articulation	Root speed	FK-pose gap (bl)
Buffalo	AnyTop (8)	0.83	0.55	0.30	–
	Ours, zero (5)	3.32	1.45	0.43	0.323
	Ours, LoRA (5)	0.33	0.30	0.44	0.132
Gazelle	AnyTop (8)	0.86	0.85	0.16	–
	Ours, zero (2)	13.2	5.39	2.13	0.255
	Ours, LoRA (2)	0.86	0.77	0.21	0.084
Dragon	AnyTop (8)	0.70	0.50	0.26	–
	Ours, zero (3)	2.53	1.02	1.44	1.215
	Ours, LoRA (3)	0.52	0.49	0.47	0.146
Spider	AnyTop (8)	0.41	0.48	n/a	–
	Ours, zero (6)	6.49	2.87	n/a	1.029
	Ours, LoRA (6)	0.20	0.24	n/a	0.130

library model has never seen these rigs: the adapters train on the rig’s own support clips, and the Truebones-only model trains on all of its captioned clips, evaluation clips included. AnyTop is unconditional, so a text-conditioned comparison under our retrieval protocol is not defined; we do not retrain it, and no rig is unseen by both models, so the numbers in Table 7 describe what a model that trained on these rigs produces, next to our own zero-training and adapted arms on the same rigs, and are not a ranking. We sample the released weights eight times per rig (6 s each) and report, for its direct output positions and for the BVH it exports after an inverse-kinematics fit, the same per-rig statistics as for our arms. These are distribution-level descriptions of how much and how smoothly the outputs move relative to the rig’s real clips; they are not a fidelity or text-alignment measure, and a lower jitter or root speed can mean less motion rather than better motion. AnyTop’s IK export raises its jitter by 1.0–1.7 $\times$ over its direct positions, whereas the pose and FK rows of our adapters have similar jitter (0.33 vs. 0.32 on Buffalo); the un-adapted library model is 1.3–1.8 $\times$ jerkier through its rotations than through its positions, which the adapter removes.

Protocol details. AnyTop samples are produced with the authors’ released model and sampling script; their BVH files declare 24 fps but the generator runs at an effective 20 fps, and we use the effective rate. Our samples use the deployment sampler (20 Euler steps, $s=2$, seed 7, rest-frame demonstration) and are read in both families: the directly predicted positions and the forward kinematics of the predicted rotations on the rig’s own bone lengths; the table reports the kinematic family, which is the deployable output, and the gap between the two is the FK-pose gap of §4.1. Joint order is validated against the rig’s conditioning table before any metric is taken. All sequences, real and generated, are decimated to a common 20 Hz with an anti-aliased polyphase filter when an AnyTop sample is present, and kept at the native 30 Hz otherwise. Ratios are means over the rig’s clips; where a rig has three or more clips we report 95% bootstrap intervals in the artifacts.

H EVALUATOR CONTROLS

The retrieval numbers come from an evaluator we trained, so we measure what that evaluator responds to. Every arm’s saved samples are re-scored in five variants with the frozen evaluator, the same pools of 64 in the same dataset order, and the same acceptance sets. *FK* keeps the sample’s rotations, root track, heading and contacts and replaces its joint positions and velocities by the forward kinematics of those rotations, so it asks whether the rotations carry the same motion the positions do. *Shuffle* permutes the sample’s frames inside the scored window. The other two are not the model’s output at all: *static* is the target clip’s own first pose held for the clip’s duration with zero velocity, and *rest* is the rig’s rest pose held the same way – degenerate sequences built from real data, which set the floor a generated sequence has to clear. *Wrong caption* gives every slot in a pool a different

clip’s caption, chosen so that no slot keeps its own caption text whenever the pool’s duplicates allow it. The evaluation is bound to the scoring it validates: the canonical pass records the digest of the scored clip ids in encoding order, and a control run is refused unless it reproduces that digest, the generation plan, and the canonical R-precision of the generated samples.

Table 8: Evaluator controls: R@1 in pools of 64 over the 3,899 validation clips. The 95% bootstrap interval over the 60 pools is 0.029 wide for the generated column and at most 0.047 wide for any other ($0.922 \in [0.907, 0.936]$ for the 303M model). The static and rest columns are built from real clips rather than from the model, and come out identical for every arm, as re-scoring the same pools with the same evaluator should.

Arm	generated	FK	shuffle	static	rest	wrong caption
TopX (303M)	0.922	0.923	0.805	0.346	0.109	0.005
36M, KTJD-17 per-cell	0.864	0.859	0.755	0.346	0.109	0.010
36M, AnyTop-13 layout	0.871	0.836	0.749	0.346	0.110	0.007
36M, scale-only	0.843	0.831	0.745	0.346	0.109	0.011
real clips	0.964			chance	0.018	

Diversity. On the same samples, the mean distance between the embeddings of two lists of 300 clips drawn without replacement is 1.4035 for the 303M model and 1.3986 for the 36M one, against 1.4035 for the real clips of the same set: neither model has collapsed onto a smaller set of motions than the data holds. The statistic is bounded by 2 here because the evaluator normalises its embeddings, and both real and generated clips sit near $\sqrt{2}$, the value of mutually orthogonal directions; it therefore separates a collapsed generator from a healthy one and says little between two healthy ones, and it is not comparable with values published for other evaluators. We do not report multi-modality: it needs its own generation pass, whose provenance our sampler does not record at the same standard as the frozen protocol.

Two readings matter. The FK column is the mechanism: re-encoding the sample through forward kinematics of its own rotations costs 0.005 of R@1 on KTJD-17 and 0.035 on the 13-channel layout, at the same model size, partition and epoch – the two redundant channels of KTJD-17 carry the same motion, those of the 13-channel layout do not, which is what the geometry of Table 11 measures directly. On the 303M model the re-encoded sample scores 0.001 above the direct one, inside the interval. The static column is the shortcut: a body of the right species, frozen in the clip’s first pose, already retrieves its caption in a third of the pools, which is the part of every retrieval number in this paper that is not motion. Uncertainty for the FID is reported as the FID between two halves of the real validation set (0.015 ± 0.001 over 20 splits), which is the estimator’s resolution at half the sample size, rather than as a bootstrap interval: FID is a biased nonlinear estimator and its percentile interval need not contain the population value.

I ABLATION NOTES

Table 9: Ablations on the 36M pilot at matched optimiser steps (epoch-200 checkpoints; validation loss, FK–pose gap, R@1 and FID on the validation set). The variant without the acceleration term optimises a different objective, so its validation loss is not listed.

Variant	Val. loss↓	FK–pose gap (bl)↓	R@1↑	FID↓
Full recipe (rest demo, calibrated, x-pred, qk-norm)	0.247	0.171	0.878	0.011
Without the acceleration-matching term	–	0.171	0.877	0.011

Prediction target and sampler. Predicting $\mathbf{x}_1$ rather than the velocity lowered the high-frequency energy of the samples in our measurements, and we deliberately keep the uniform t sampler: the logit-normal sampler and $\sigma_{\min}=0.05$ of Li & He (2025) converged faster in loss but concentrated nearly all supervision at $t \geq 0.8$ and visibly degraded global form, so they were rolled back on visual grounds.

Acceleration-matching term. The pilot trained without the acceleration term of §3.3 reaches the same FK–pose gap (0.171 bone lengths), R@1 (0.877 against 0.878) and FID (0.011) at epoch 200 as the full recipe; the term was introduced for temporal smoothness, which these three numbers do not measure.

Normalisation of the representation. KTJD-17 is normalised per rig and per cell (joint, channel) by the cell’s mean and standard deviation over the rig’s clips (App. A). The alternative that keeps every cell’s own mean and divides only by a per-rig scale, as in the KTJD specification, was trained with the 36M recipe unchanged and scored at matched epochs under the frozen protocol (its samples are mapped into the per-cell space before scoring). Table 10 shows that the scale-only variant is behind in retrieval and FID at every matched epoch, by about two points of R@1 and 22–30% of the FID, although its two output channels agree more closely with each other: the FK–pose gap is computed after each arm’s own normalisation is reversed, in the same bone-length unit, so the scale-only model’s smaller gap says that its two channels are more consistent with each other, not that its samples are closer to the target – retrieval and FID say the opposite. The per-cell normalisation is the one used everywhere else in the paper.

Table 10: Per-cell against scale-only normalisation at matched epochs (36M recipe, global batch 128, frozen protocol: pool 64, 20 Euler steps, guidance 2, seed 42). FK–pose gap in bone lengths from the training-time validation pass at the same epoch.

Epoch	Per-cell mean/std (ours)			Scale-only		
	R@1↑	FID↓	FK–pose gap↓	R@1↑	FID↓	FK–pose gap↓
50	0.852	0.0157	0.198	0.832	0.0201	0.117
75	0.860	0.0146	0.193	0.840	0.0194	0.111
100	0.864	0.0138	0.187	0.843	0.0198	0.107

Representation content. To separate what the channel layout of KTJD-17 contributes from what its normalisation contributes, the older 13-channel AnyTop layout (root-relative, heading-canonical joint positions, the root’s height and planar velocity, per-parent local 6D rotations, local velocities and contacts; no root-track or heading channels) was derived analytically from the same clips, splits and exclusion cut, and the 36M recipe was trained on it with the same per-cell normalisation and the same schedule. The two world-coordinate terms of the training loss (forward-kinematic position and foot lock) are defined on KTJD-17’s world coordinates and have no counterpart in the 13-channel layout, so that run had them off; the velocity and acceleration terms were kept. Its samples are converted back to KTJD-17 before scoring, so the frozen protocol and the reference numbers are unchanged. Table 12 shows the two layouts within a point of R@1 of each other at every matched epoch (0.861, 0.864 and 0.871 against 0.852, 0.860 and 0.864), with the FID differing by at most 0.002: the retrieval gains in Table 10 come from the normalisation, not from the channel layout.

That comparison changes the objective as well as the layout, so we ran KTJD-17 with those two terms off, at the 13-channel arm’s own batch partition. At epoch 50 it reaches R@1 0.868 and FID 0.0140, against 0.861 and 0.0154 for the 13-channel layout under the same objective and partition: with the objective held fixed the two layouts are 0.007 of R@1 apart, and the layout is not what the retrieval numbers are measuring. Read the other way, the same two terms cost about a point and a half of R@1 on KTJD-17 (0.868 without them against 0.852 with them, across a change of batch partition) and buy the geometry of Table 11: without them the generated rotations and positions are free to disagree, which is exactly what the 13-channel layout’s FK–pose gap of 1.447 bone lengths shows. The 13-channel layout has no rotation slot for leaf joints, whose orientation the conversion fills from the parent (about ten degrees of mean error on ground truth); the layout used in the paper keeps it. What the layout does change is measured on the same frozen samples (Table 11): every generated clip and its ground-truth clip are decoded with the official codec and compared in the rig’s mean bone length. The two decodes of a KTJD-17 sample, its positions and the forward kinematics of its rotations, agree to 0.38 bone lengths against 1.45 for the 13-channel layout; its direct positions keep the rest bone lengths better; its leaf joints articulate (10.6 degrees of mean deviation from rest against 11.6 degrees in the data, where the 13-channel layout has no leaf rotation to articulate); its feet slide less while flagged in contact (2.3 against 3.8 bone lengths per second, the data 2.0); and its root travels and stands where the data does, within a few percent in displacement and height, against

22% further and 4% lower for the layout that integrates a root velocity. The 13-channel layout is a little smoother at the root and articulates internal joints a little more. These are the quantities the layout was designed around; retrieval does not resolve them.

Table 11: Geometry of the generated clips against their ground truth (epoch 100, the 3899 validation clips and the same frozen samples as Table 12; official codec; bl = the rig’s mean rest bone length). Last column: share of clips on which the KTJD-17 sample is closer to the ground truth than the 13-channel sample (ties count for neither). Every row uses all 3899 clips except foot slide, which each stream computes under its own contact flags and which is therefore compared only on the 3610 clips where all three are defined. Epochs 50 and 75 give the same picture.

Quantity	Data	KTJD-17	AnyTop-13	KTJD-17 closer
FK-pose gap (bl)	0.000	0.381	1.447	100%
Bone-length error, direct positions (bl)	0.000	0.077	0.119	91%
Leaf articulation, mean deviation from rest (deg)	11.6	10.6	0.0	92%
Leaf articulation, temporal std (deg)	5.86	5.24	0.00	89%
Foot slide while in contact (bl/s)	2.02	2.32	3.80	71%
Root displacement (bl)	8.87	9.24	10.83	61%
Root path length (bl)	11.26	11.04	12.87	55%
Root height (bl)	6.97	7.05	6.71	54%
Heading turn (rad)	1.07	1.13	1.04	43%
Root jitter (bl)	0.038	0.047	0.044	30%
Joint jitter (bl)	0.094	0.090	0.112	49%
Internal-joint articulation, mean deviation from rest (deg)	19.3	17.9	18.8	42%
Internal-joint articulation, temporal std (deg)	8.99	7.11	8.16	32%

Table 12: Channel layout at matched epochs (36M recipe, global batch 128, per-cell normalisation in both; frozen protocol: pool 64, 20 Euler steps, guidance 2, seed 42). The AnyTop-13 rows were sampled with TF32 matrix multiplication, the KTJD-17 rows in fp32.

Epoch	KTJD-17 (ours)		AnyTop-13 layout, derived	
	R@1↑	FID↓	R@1↑	FID↓
50	0.852	0.0157	0.861	0.0154
75	0.860	0.0146	0.864	0.0161
100	0.864	0.0138	0.871	0.0155

Model size. Table 13 scores three sizes of the same recipe – width, depth and head count grow together, so it measures model size and not the contribution of any one of them – at the same number of optimiser steps and data passes (epoch 200, global batch 128, frozen protocol). Retrieval and FID improve monotonically with size; the 303M model at epoch 200 is within a point of R@1 of its epoch-289 checkpoint in Table 1 (0.914 against 0.922), so most of its gain over the pilot comes from size, not from the longer schedule.

Table 13: Model size at matched optimiser steps (epoch 200, global batch 128; frozen protocol, pool 64, 20 Euler steps, guidance 2, seed 42). The 36M row was sampled in fp32, the two larger rows with TF32 matrix multiplication.

Model	Width / depth / heads	R@1↑	R@3↑	FID↓
36M	384 / 8 / 6	0.878	0.976	0.0113
116M	640 / 10 / 10	0.900	0.985	0.0071
303M	896 / 14 / 14	0.914	0.990	0.0052

Stability. Without query-key normalisation the 303M model diverged in every attempt, between epochs 11 and 60, whichever of learning rate, gradient clipping, spike rejection or batch size was changed; probing the attention logits of a healthy and of a collapsing checkpoint of one such run showed the first block’s temporal attention already saturated on the healthy checkpoints (maximum

logit 1372–1483, where a sane transformer runs at tens) and exploding to 12392–23391 across the damaging step. With query–key normalisation the same recipe trained for 290 epochs without a single rejected step.

J ADDITIONAL QUALITATIVE RESULTS

Figure 3 shows the zero-training and LoRA arms of Table 2 on one held-out clip per rig. Every panel overlays the two decodings of the same sample: the directly predicted joint positions and the forward kinematics of the predicted rotations. The zero-training arm produces two skeletons that disagree on every rig, most on the Dragon and the Spider, whose gap exceeds a bone length; the adapter brings them into agreement, which is what the drop of the FK–pose gap in Table 2 measures.

K VALIDATION SPLIT

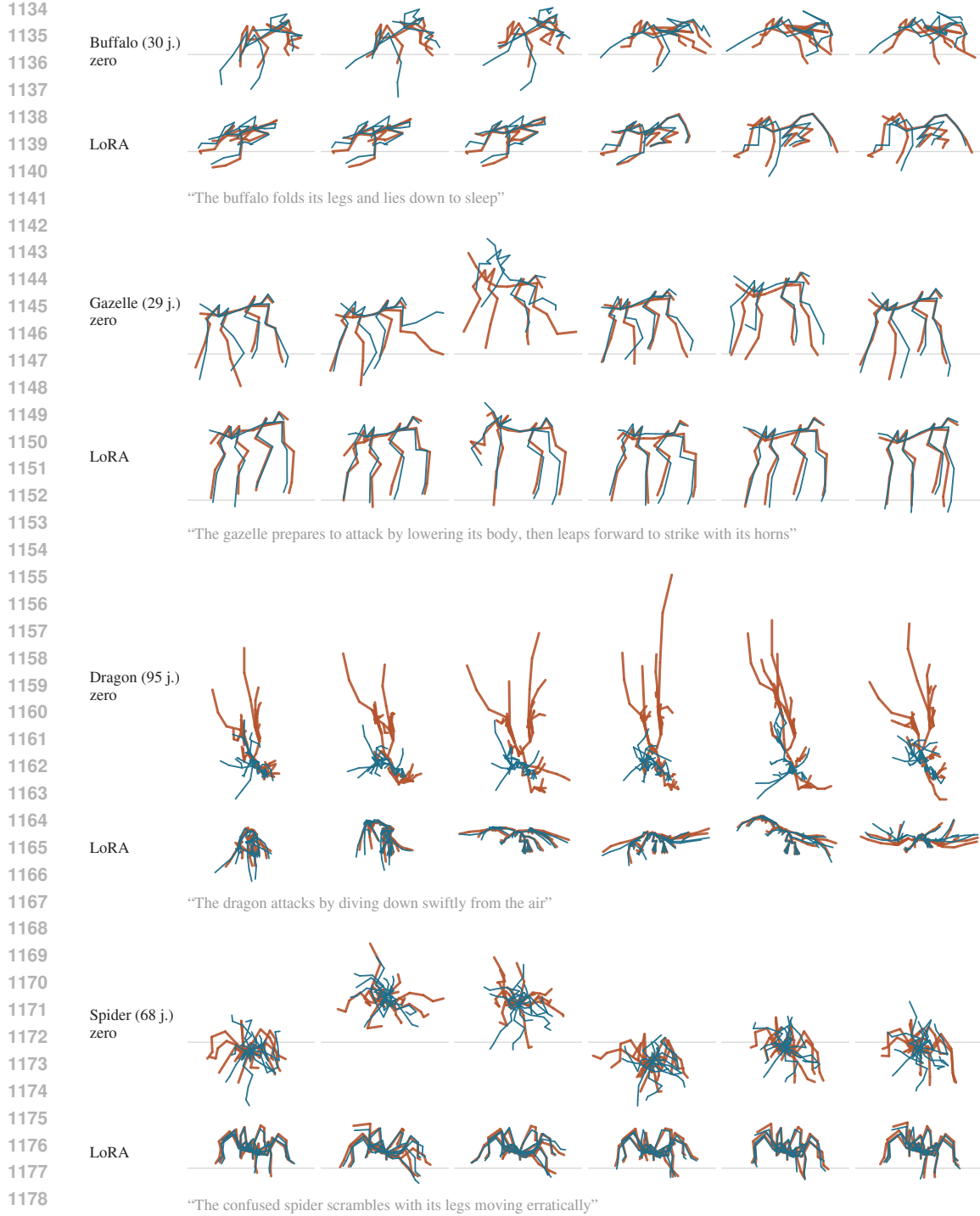
The library is split per rig: the rig’s clips are sorted by identifier and permuted with one seeded generator (numpy default generator, seed 0) shared by all rigs in name order, and the first 5% of the permutation (at least one clip, none for a rig with a single clip) become validation, so that every rig with enough clips is represented in both parts. The split is fixed before any model is trained and is shared by the evaluator and by every generator and adapter in this paper; checkpoints are selected on the validation loss. Validation holds 3,899 clips over 311 rigs and training 73,995 clips. Retrieval pools are formed by chunking the validation set in dataset order (rigs sorted by name), so a pool of 64 holds clips of one rig or of neighbouring rigs; the last 59 clips do not fill a pool and are excluded from the R-precision but not from the FID or the matching score.

Shared source animations. The annotation set exports one source animation to several rigs, so a motion can appear under more than one clip; retrieval accounts for this in its acceptance sets, which hold every export of a caption’s source motion. On the 512 validation clips over 216 rigs whose motion payload and (rig, source animation) pair appear nowhere in training, the 303M model reaches R@1 0.967 against 0.918 on the full validation set at the same pool size of 32, with the real clips at 0.988 and 0.975.

Pool size. Retrieval among more candidates is harder for the real clips and for the generated ones alike, and the pool size is a free parameter of the protocol. Table 14 re-scores the same generated samples at four pool sizes; the ranking of the two models does not depend on the choice, while the gap to the real clips widens with the pool (from 0.030 at 16 to 0.046 at 128 for the 303M model); the main text uses 64.

Table 14: R@1 of the same samples (20 steps, $s=2$, one seed) re-scored in pools of 16, 32, 64 and 128 formed by chunking the dataset order; the incomplete last pool is dropped. The real clips are scored by the same evaluator in the same pools.

Pool size	16	32	64	128
Pools	243	121	60	30
Real clips (reference)	0.985	0.975	0.964	0.961
TopX (pilot, 36M)	0.938	0.918	0.897	0.891
TopX (303M)	0.956	0.937	0.922	0.915



1181 Figure 3: Truebones arms of Table 2: one held-out clip per rig, six frames spread evenly over the
 1182 clip, three-quarter view; the quoted sentence is the conditioning prompt. In every panel the directly
 1183 predicted joint positions (clay) are overlaid by the forward kinematics of the predicted rotations
 1184 (teal); a visible separation is the projected FK-pose discrepancy (a single view can hide a disagree-
 1185 ment in depth). Rows: the library model with the rig’s statistics only (*zero*, the unadapted snapshot,
 1186 whereas the zero column of Table 2 is read after at most 15 warm-up steps) and the per-rig adapter
 1187 (*LoRA*). Frames are anchored on the root horizontally and keep their height above the floor; the grey
 line marks the floor height beneath the root. The Dragon, a flying rig, is anchored on the root in all
 axes, which removes its dive trajectory. Both rows of a rig are drawn at the same scale.